PhysWM: Grounding Predictive World Models in Physical Equations

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Abstract

Predictive world models can forecast dynamics without organizing their representations around the physical variables governing a system. We introduce PHYSWM, a two-path architecture that grounds an action-conditioned predictive world model through known physical equations. A neural path predicts the next state from data, while an equation path maps the same predictive latent through a low-capacity probe to physical variables and a frozen differentiable solver. Without parameter labels, the equation path learns to reproduce the neural path's stopped-gradient prediction. We evaluate the method on FetchPush, randomized PushT, and CartPole. On FetchPush, the neural path beats persistence in every seed and equation grounding reduces teacher-matching error from 8.758 to 0.752. On randomized PushT, inferred effective dynamics beat shuffled episode assignments in all three seeds. FetchPush exposes a solver-adequacy failure, while the small CartPole runs remain below the data floor. These results support equation-compatible effective coordinates without equating them with uniquely recovered material parameters.

1 Introduction

World models compress observations and actions into representations useful for predicting future dynamics (4; 5; 6; 11). Predictive accuracy alone, however, does not ensure that these representations expose mass, damping, stiffness, friction, or control response. A high-capacity predictor may exploit trajectory correlations while leaving the governing physical factors entangled, even when its next-state predictions are accurate (9). This separates predicting *what happens* from representing *why the system evolves that way*.

We study whether known physical equations can ground a flexible predictive world model without physical-parameter annotations. PHYSWM produces two next-state predictions from the same action-conditioned latent (Figure 1). Path A is a learned decoder trained on observed transitions. Path B maps the latent to a compact vector $\hat{\theta}$ and evaluates a frozen differentiable solver. Path B targets Path A's stopped-gradient prediction rather than the dataset state directly. The neural path remains anchored to data and free to absorb model mismatch; the physical path asks whether the model's predictive belief admits a compact equation-based explanation. This fixed-decoder role is related to differentiable system identification (8), but our training target is the predictor's belief rather than rendered pixels or labeled physical parameters.

Our contribution is threefold. First, we introduce a shared-latent, two-path objective for grounding predictions in a supplied equation family without parameter labels. Second, we evaluate grounding functionally by changing only the solver's parameter vector while holding states, actions, weights, and equations fixed. Third, we distinguish *semantic parameters*, which match named ground-truth quantities, from *effective physical coordinates*, which improve equation rollouts but may compensate for non-identifiability or solver mismatch.

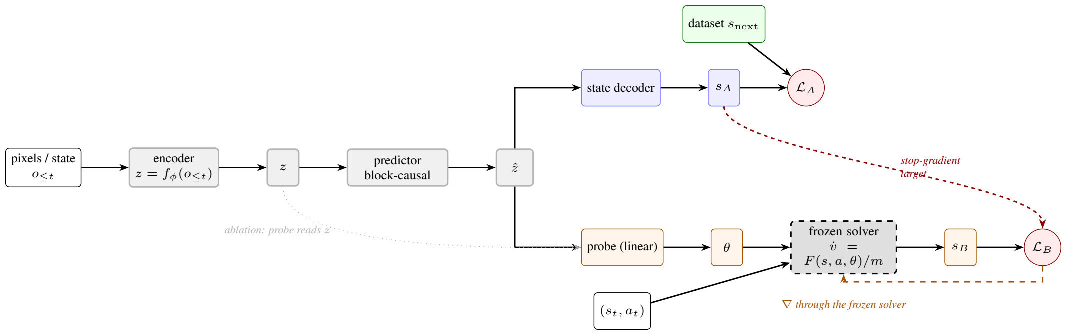

Figure 1: **PHYSWM architecture.** Encoder and predictor form one shared trunk producing a single action-conditioned latent $\hat{z}$; both heads read $\hat{z}$, never a private view of the observation. Path A (blue) is the ordinary predictor: a decoder reads $\hat{z}$ to a state s_A, and $\mathcal{L}_A$ (Equation (4)) regresses the dataset's ground truth, exactly what a predictive world model already does. Path B (orange) reads a physics vector θ off the same $\hat{z}$ with a deliberately low-capacity probe, integrates it through a *frozen, zero-parameter* solver (gray, dashed) to get an independent estimate s_B. $\mathcal{L}_B$ does *not* use the dataset label: it regresses s_A with gradient stopped (red, dashed), so the only route back to a learnable parameter is through the fixed solver into θ (orange dashed arrow), and θ can only improve by becoming a better physical explanation of what Path A already predicts. The dotted arrow is not part of the method: it marks the pre-action routing ablation (Section 3), where the probe reads z instead of $\hat{z}$.

2 Equation-grounded predictive representations

Given observations o_t, actions a_t, and prediction state s_t, an encoder and causal action-conditioned predictor form

$$z_t = E_\phi(o_t), \qquad \hat{z}_t = P_\psi(z_{\leq t}, a_{\leq t}). \tag{1}$$

The neural decoder produces $\hat{s}^A_{t+1} = D_\omega(\hat{z}_t)$ and is trained in normalized state space:

$$\mathcal{L}_A = \|N(\hat{s}^A_{t+1}) - N(s_{t+1})\|_2^2. \tag{2}$$

A bounded low-capacity probe pools only causal context latents,

$$\hat{\theta} = \mathrm{Bound}(\rho_\xi(\mathrm{Pool}(\{\hat{z}_t\}_{t\in\mathcal{C}}))), \tag{3}$$

and a frozen solver S, containing no learnable parameters, produces $\hat{s}^B_{t+1} = S(s_t, a_t, \hat{\theta})$. The grounding loss is

$$\mathcal{L}_B = \|N(\hat{s}^B_{t+1}) - \mathrm{sg}(N(\hat{s}^A_{t+1}))\|_2^2, \qquad \mathcal{L} = \mathcal{L}_A + \alpha\mathcal{L}_B. \tag{4}$$

The stopped target prevents $\mathcal{L}_B$ from moving the Path-A decoder through its target edge, while gradients through S, ρ_ξ, and P_ψ shape the shared predictor. The default probe is linear, preventing it from hiding another world model. $\hat{\theta}$ is always a forward-pass output, never a free test-time optimization variable. The stop-gradient follows self-distillation methods (7; 1; 3); here it enforces a one-way dependency between a predictive teacher path and an equation-constrained student path.

Causal evaluation split. The identifier observes only completed context transitions. Metrics are computed on later query transitions, and no query outcome is accessible to the probe. With the default eight-frame window, four completed transitions form the context. Functional evaluation uses eight context transitions followed by seven held-out rollout transitions.

Grounding criterion. We call a coordinate physically grounded when reading it from the predictive latent allows the fixed equations to explain held-out dynamics better than a coordinate carrying no episode-specific information. This is deliberately weaker than claiming recovery of a unique material parameter: trajectories may identify only ratios such as stiffness/mass, and approximate solvers may induce compensating effective values.

(a) FetchPush (b) PushT (c) CartPole

Figure 2: Rendered observations from the three evaluated environments. FetchPush and CartPole use MuJoCo; PushT uses Pymunk contact dynamics.

Table 1: Mean $\pm$ sample standard deviation over three seeds. RMSE is lower better; R^2 is higher better.

Benchmark	Metric	Baseline / control	PhysWM
FetchPush	Prediction RMSE	Persistence 0.648 ± 0.037	0.420 ± 0.015
FetchPush	Teacher-match RMSE	Ungrounded 8.758 ± 0.398	0.752 ± 0.036
PushT	Substitution RMSE	Shuffled 0.267 ± 0.013	Inferred 0.208 ± 0.011
CartPole	Substitution RMSE	Shuffled 0.339 ± 0.107	Inferred 0.311 ± 0.155
CartPole	Prediction RMSE	Persistence 0.199	0.792

3 Evaluation

We test four complementary properties. (1) *Predictive adequacy*: Path A must beat persistence, $\hat{s}_{t+1} = s_t$. (2) *Semantic readout*: where labels exist, a supervised linear probe measures held-out episode-level R^2. (3) *Parameter substitution*: the solver receives the inferred vector, another episode's shuffled vector, nominal constants, or true parameters when available. (4) *Multi-horizon rollout*: the same substitutions are rolled forward under fixed held-out actions. All errors are scaled per state dimension.

FetchPush. We use the Gymnasium-Robotics FetchPush task with per-episode object mass and friction variation. The MuJoCo environment supplies real contact dynamics, while the frozen solver uses a point-mass, linear-friction approximation. The reported primary runs use 256 episodes and three seeds.

Randomized PushT. This planar manipulation benchmark extends the contact-rich Push-T task (2) with episode-level variation in effective dynamics. Its solver is approximate and ground-truth labels are unavailable for most coordinates, making inferred-versus-shuffled substitution the primary test. The reported runs use 512 episodes and three seeds.

CartPole. The DeepMind Control Suite task (10) uses a six-parameter analytic solver for cart mass, pole mass, pole half-length, gravity, force gain, and damping. Only small, independently sampled runs of 8, 32, and 64 episodes are complete, so we treat CartPole as a pipeline and data-floor diagnostic rather than headline evidence.

4 Results and discussion

FetchPush satisfies the predictive premise. Path A reaches 0.420 ± 0.015 scaled RMSE, compared with 0.648 ± 0.037 for persistence, and wins in every seed (Table 1). The physics loss also changes the equation path substantially: its error against the stopped-gradient teacher falls from 8.758 ± 0.398 without grounding to 0.752 ± 0.036 with grounding. The shared latent therefore supports an equation-

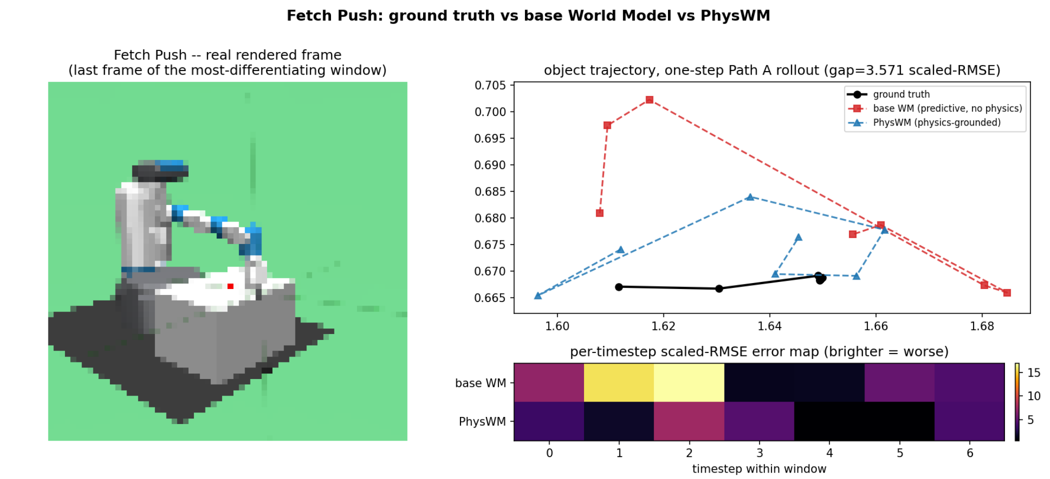

Figure 3: FetchPush error map on the held-out window with the largest difference between the two Path-A models. Left: the real final frame. Top right: ground-truth object trajectory against the predictive baseline and PhysWM. Bottom right: per-timestep scaled state RMSE; brighter cells indicate larger error. The models predict state, not pixels, so the rendered frame supplies visual context rather than a pixel-reconstruction target.

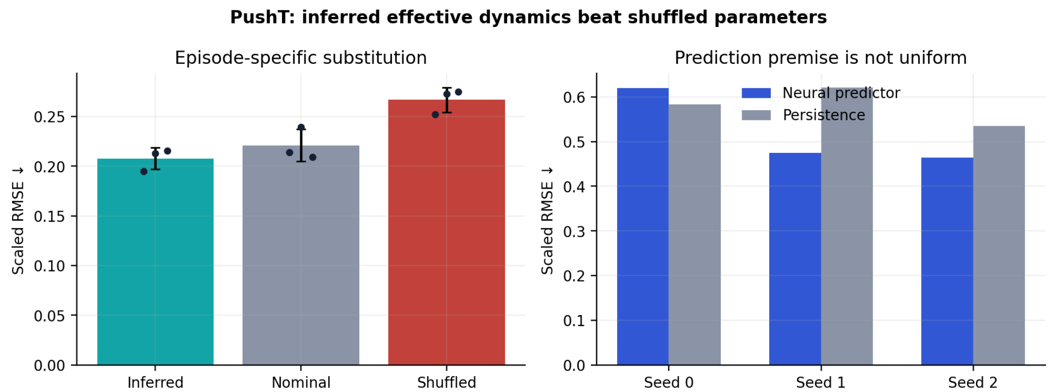

Figure 4: Randomized PushT substitution test over three seeds. Inferred episode-specific coordinates beat shuffled assignments in every seed and nominal parameters in two of three. Error bars show sample standard deviation.

constrained reconstruction of the neural prediction, even though the raw mass and friction coordinates are not semantically identifiable at this data budget.

PushT coordinates carry episode-specific dynamics. Inferred variables obtain 0.208 ± 0.011 substitution RMSE versus 0.267 ± 0.013 for shuffled assignments (Fig. 4). They beat shuffled values in all three seeds and nominal values in two of three. Solver sensitivity is concentrated in the agent's proportional and derivative gains, so this supports effective controller dynamics rather than complete contact-parameter identification. Path A fails to beat persistence in one seed, which limits the strength of this result.

FetchPush coordinates compensate for solver mismatch. The corrected frozen solver fails its adequacy check even after directly fitting a free parameter vector to each episode: fitted solver RMSE is 0.483, compared with 0.411 for persistence. In the functional run, inferred variables produce lower error than true mass and friction (1.535 versus 6.984), while solver-space task success is 0.03125 for every parameter source. The probe is compensating for the solver's fixed contact and point-mass approximations. We therefore call $\hat{\theta}$ an effective coordinate rather than a recovered material parameter.

CartPole is still data-limited. Across independent 8-, 32-, and 64-episode runs, inferred coordinates beat shuffled coordinates in a majority of seeds at every scale, but neither Path A nor the inferred solver beats its naive baseline. At 64 episodes, probe and shuffled substitution errors are 0.311 ± 0.155 and 0.339 ± 0.107, while nominal parameters reach 0.007. The non-monotonic Path-A error across budgets rules out a scaling claim from these small runs.

5 Conclusion and limitations

PHYSWM grounds a predictive world model by requiring its shared action-conditioned representation to support a compact explanation through known physical equations. FetchPush establishes the predictive premise and shows that grounding makes the equation path track its teacher. Randomized PushT shows episode-specific information under parameter substitution. Neither result establishes recovery of uniquely named material parameters.

The study is limited to tiny-CNN diagnostics. The FetchPush solver is inadequate even after oracle fitting, PushT uses an approximate disc model for a T-shaped block, and CartPole has only small independent samples. The method does not discover equations, select informative actions, or demonstrate closed-loop planning or sim-to-real transfer. Larger CartPole runs, stronger visual encoders, and solvers that pass adequacy checks are required before broader claims.

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Closest prior work in spirit: aligns latent states with physical quantities and constrains their evolution with a partially-known dynamics model, without ground-truth physical annotations. The supervision differs in kind: PIWM uses weak, distribution-based physical labels and a learnable dynamics model; our training signal uses no physical label at all, and the dynamics model is frozen rather than learned.

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